

Mid Semester Exam
DA526 Image Processing with Machine Learning (Jan-Apr 2026)

Date (Time): 03-MAR-2026 (09:00 – 11:00)

Maximum Marks: 100

Name: _____

Roll No.: _____

Instructions:

- (1) There are a total of nine (9) questions. All questions are compulsory.
- (2) Show complete steps for all derivations. Clearly state assumptions wherever necessary.
- (3) You must submit the question paper along with the answer sheets.
- (4) Ensure that your name and roll number are written on both the question paper and the answer sheets.

Q1 [10 marks] Define the following terms with mathematical expressions:

- | | |
|---------------------------|---------------------------------|
| (a) Machine Learning | (f) Loss Function |
| (b) Supervised Learning | (g) Parameter vs Hyperparameter |
| (c) Unsupervised Learning | (h) Dimensionality |
| (d) Generative Model | (i) Normalization (Min-Max) |
| (e) Discriminative Model | (j) Standardization (Z-score) |

Q2 [10 marks]

- (a) Write the linear regression model without bias in vector form.
- (b) Define residual and write the Least Squares objective function.
- (c) Express the objective function in matrix form.
- (d) Derive the gradient of $f(w) = \frac{1}{2} \|Xw - y\|^2$.
- (e) Starting from $f(w) = 0$, derive the normal equation.
- (f) Obtain the least squares solution for w .
- (g) State the condition under which the solution exists.

Q3 [10 marks]

- (a) Explain the effect of increasing polynomial degree p .
- (b) Define High Bias and High Variance.
- (c) Explain Underfitting and Overfitting.
- (d) Why does higher polynomial degree not always mean better model?

Q4 [15 marks] Consider the following two imaging systems:

System A: High spatial resolution, low intensity resolution

System B: Low spatial resolution, high intensity resolution

- (a) Mathematically define spatial resolution and intensity resolution.
- (b) For medical imaging, which system would be preferable? Justify analytically.
- (c) Explain how quantization error depends on intensity resolution.
- (d) Derive quantization step size for an 8-bit and 3-bit system over range $[0, 255]$.

Q5 [10 marks] The basic image formation model is:

$$f(x, y) = i(x, y) \cdot r(x, y)$$

where $i(x, y)$ is illumination and $r(x, y)$ is reflectance.

- (a) Explain physical interpretation of each term.
- (b) Show how logarithmic transformation converts multiplicative model into additive form.
- (c) Explain why this transformation is useful in image enhancement.



a b c

FIGURE 2.25 (a) Image with a low level of detail. (b) Image with a medium level of detail. (c) Image with a relatively large amount of detail. (Image (b) courtesy of the Massachusetts Institute of Technology.)

Q6 [10 marks] The three images shown in Figure 2.25 (a), (b), and (c) show scenes with increasing levels of detail. If each image is 512×512 pixels.

- (a) Explain why image (c) may require more complex modeling.
- (b) Discuss curse of dimensionality in context of high-detail image classification.
- (c) Suggest a dimensionality reduction approach and justify.

Q7 [15 marks] The three images shown in Figure 2.25 (a), (b), and (c) represent scenes with increasing levels of detail. Assume each image has intensity levels in the range $[0, 255]$. Consider the general intensity transformation:

$$g(x, y) = T(f(x, y))$$

- (a) If a negative transformation is applied, explain how the visual appearance of each image (a), (b), and (c) would change. Which image would show the most perceptual difference? Justify.
- (b) Suppose a logarithmic transformation is applied, discuss which of the three images would benefit most from this transformation. Explain your answer in terms of dynamic range and fine details.
- (c) Now consider a power-law (gamma) transformation is applied:
 - (i) If $\gamma < 1$, which image would show more visible enhancement in darker regions?
 - (ii) If $\gamma > 1$, which image would lose more shadow detail?
 Note: Provide reasoning based on intensity distribution.
- (d) If a contrast stretching transformation is applied that expands the intensity range from $[80, 160]$ to $[0, 255]$, which image is most likely to show significant contrast improvement? Explain.

Q8 [10 marks] Given the following 4×4 grayscale image (3-bit depth):

$$\begin{bmatrix} 3 & 3 & 4 & 5 \\ 2 & 3 & 4 & 6 \\ 2 & 2 & 4 & 6 \\ 1 & 2 & 5 & 7 \end{bmatrix}$$

- (a) Compute the histogram.
- (b) Compute normalized histogram.
- (c) Compute cumulative distribution.
- (d) Perform histogram equalization and obtain the transformed image.

Q9 [10 marks] An image of size 100 pixels has intensity values distributed as: 60 pixels at level 2, 30 pixels at level 3, and 10 pixels at level 7 (Assume $L = 8$).

- (a) Perform histogram equalization.
- (b) Compute new intensity mapping.
- (c) Compare dynamic range before and after equalization.
- (d) Explain why HE improves contrast in low-contrast images.